

**Draft - expert review pending**

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

# Unit 4 assessment

## Unit summary

Every measurement is a number-unit pair on a ladder: length in metres, mass in kilograms, capacity in litres, with kilo-, centi- and milli- stepping by powers of ten - convert before computing, and never report a bare number. Perimeter is the boundary in metres, area the cover in square metres, volume the fill in cubic metres or litres; the rectangle, triangle, circle and box formulas carry pictures (the half, the pi), not spells, and area and perimeter move independently. A real measurement task runs choose, measure, compute, check - the quantity the task buys names the unit, and a round-number estimate catches the impossible answers - and the four moves written out are the report a colleague can re-check.

## Summative assessment

### Multiple-choice questions (MCQs)

1. 2.5 metres, converted to centimetres, is: (*Remember*) a) 25 cm b) 250 cm c) 2,500 cm d) 0.25 cm
2. "The tank holds 500" is not a measurement because: (*Understand*) a) tanks are never measured b) the number lacks its unit c) 500 is too large for a tank d) capacity is measured in kilograms
3. A roll of cloth is 250 cm long. Rolls needed for 15 metres: (*Apply*) a) 6 b) 17 c) 4 d) 60
4. The perimeter of a rectangle 8 m by 6 m is: (*Apply*) a) 48 m b) 14 m c) 28 m d) 16 m

5. The area of the same rectangle (8 m by 6 m) is: (*Apply*) a) 48 square metres  
b) 28 square metres c) 14 square metres d) 480 square metres
6. In the triangle area formula, the divide-by-two exists because: (*Understand*)  
a) triangles have three sides b) a triangle is half of its enclosing rectangle c)  
angles sum to 180 degrees d) the base is measured twice
7. A box 2 m by 1 m by 1.5 m holds: (*Apply*) a) 4.5 cubic metres b) 3 cubic  
metres c) 3 square metres d) 450 litres
8. A corridor 2 m by 20 m and a room 6 m by 7 m: (*Apply*) a) the corridor has  
more area and more perimeter b) the room has more area and more  
perimeter c) the corridor has more perimeter, the room more area d) the  
room has more perimeter, the corridor more area
9. Buying skirting board for a classroom is buying: (*Understand*) a) area b)  
volume c) perimeter d) capacity
10. The four moves of a real measurement task, in order, are: (*Remember*) a)  
measure, compute, choose, check b) choose, measure, compute, check c)  
compute, check, choose, measure d) choose, compute, measure, check

## Restricted-response questions (RRQs)

1. Convert: (a) 3.4 m to centimetres; (b) 5,600 ml to litres; (c) 0.9 kg to grams.  
(*Apply*)
2. Explain in one sentence why "the water tank holds 500" fails as a  
measurement, and give the two likely intended meanings. (*Understand*)
3. Compute the perimeter and the area of a classroom 9 m by 5 m, stating the  
unit of each. (*Apply*)
4. A triangular notice board has base 1.2 m and height 0.8 m. Compute its  
area, and explain where the divide-by-two comes from. (*Apply*)
5. A circular flower bed has radius 4 m. Compute its circumference and area  
(pi as 3.14). (*Apply*)
6. A wall 6 m by 3 m needs paint at 1 litre per 10 square metres. Compute the  
litres to buy, and state why the answer rounds as it does. (*Apply*)
7. A trainee reports a classroom's tile needs as 4,800 square metres. Name the  
move where the error most likely happened, and one check that would have

caught it. (*Analyze*)

8. Name the defining property of a circle, and state which length a circumference computation needs. (*Understand*)
9. A water tank is a box 1.5 m by 1 m by 1 m. Compute its volume in cubic metres and in litres. (*Apply*)
10. For each task, state what it buys and its unit: (a) lace for a dupatta border; (b) marble for a floor; (c) water for a drum. (*Understand*)

## Extended-response questions (ERQs)

1. The school kitchen prepares rice for 120 pupils at 75 grams per pupil; rice comes in 25-kilogram sacks at 4,200 rupees per sack. Compute the total rice needed in kilograms, the number of full sacks to order and any leftover, and the cost; then check the order with one estimation sentence. (*Analyze*)
2. The school wants to fence a rectangular kitchen garden 12 m by 8 m, carpet a staff room 5 m by 4 m, and install a cylindrical tank of radius 1 m and height 2 m. Compute the fencing, the carpet, and the tank's volume in litres (pi as 3.14), stating the unit of each; then check one answer with an estimation sentence and write the whole answer so a colleague could re-check every step. (*Analyze*)
3. A book corner needs a rectangular carpet 3 m by 2 m, skirting along the two walls behind it, and shelving for 250 books with 2.5 cm spines on 80 cm shelves. Compute the carpet area, the skirting length, and the number of shelves with units; then judge which quantity a "one of each" budget most likely under-planned, with a computation that supports your judgement. (*Analyze*)
4. A courtyard 30 m by 20 m has a rectangular store 5 m by 4 m jutting into it. The school wants skirting for the courtyard boundary and tiles for the open area. Compute both quantities, and analyse which of the two the school should re-measure most carefully if the store's position (but not its size) might change, with a reason. (*Analyze*)
5. Using the real measurement task you completed at your practicum school (Topic 4.3 practicum task), present its four-move plan, and evaluate it: which move was its weakest, what the cooperating teacher's verdict was, and what you would change in the plan's communication so a colleague could re-check it without asking you a single question. (*Evaluate*)

# Answers and marking guidance

## MCQ answer key

1. **b** -  $2.5 \times 100 = 250$  cm.
2. **b** - a measurement is a number-unit pair; 500 what?
3. **a** - convert first:  $15 \text{ m} = 1,500 \text{ cm}$ ;  $1,500 / 250 = 6$ .
4. **c** -  $2 \times (8 + 6) = 28$  m.
5. **a** -  $8 \times 6 = 48$  square metres.
6. **b** - a triangle is half the rectangle that encloses it with the same base and height.
7. **b** -  $2 \times 1 \times 1.5 = 3$  cubic metres (option d confuses cubic metres with litres).
8. **c** - corridor: area 40, perimeter 44; room: area 42, perimeter 26.
9. **c** - skirting runs along the boundary: perimeter.
10. **b** - choose, measure, compute, check.

## RRQ model answers and mark schemes

1. Model: (a) 340 cm; (b) 5.6 L; (c) 900 g. **Marks (3):** 1 each.
2. Model: a measurement is a number-unit pair, and the bare 500 names no unit; for a water tank the likely meanings are 500 litres or 500 millilitres - wildly different tanks. **Marks (2):** 1 - the number-unit point; 1 - both plausible units named.
3. Model: perimeter =  $2 \times (9 + 5) = 28$  m; area =  $9 \times 5 = 45$  square metres. **Marks (2):** 1 - both values; 1 - both units.
4. Model: area =  $(1.2 \times 0.8) / 2 = 0.48$  square metres; the half comes from the triangle being half its enclosing rectangle. **Marks (2):** 1 - value; 1 - the half explained.
5. Model: circumference =  $2 \times 3.14 \times 4 = 25.12$  m; area =  $3.14 \times 4 \times 4 = 50.24$  square metres. **Marks (2):** 1 each.
6. Model: area = 18 square metres; paint =  $18 / 10 = 1.8$ , so buy 2 litres; paint is sold in whole litres, so the answer rounds up. **Marks (2):** 1 - 2 litres; 1 - the rounding direction justified.
7. Model: the error is most likely at the choose-or-measure move (a unit slip such as treating centimetres as metres); a round-number check (a

classroom is about  $10\text{ m} \times 10\text{ m} = 100$  square metres, so 4,800 is impossible) would have caught it. **Marks (2):** 1 - the move named; 1 - the check that catches it.

8. Model: a circle is every point at one fixed distance (the radius) from a centre; a circumference computation needs the radius (or diameter). **Marks (2):** 1 - the defining property; 1 - the needed length.
9. Model:  $1.5 \times 1 \times 1 = 1.5$  cubic metres = 1,500 litres. **Marks (2):** 1 - cubic metres; 1 - litres.
10. Model: (a) perimeter, in metres; (b) area, in square metres; (c) volume or capacity, in litres. **Marks (3):** 1 each.

## ERQ rubrics

1. Kitchen rice. Analytic rubric: *Total* (0-2:  $120 \times 75\text{ g} = 9,000\text{ g} = 9\text{ kg}$ ); *Sacks and leftovers* (0-2: 1 sack, 16 kg left); *Cost and check* (0-2, **Analyze:** 4,200 rupees with an estimation check noting 9 kg is well under one 25 kg sack). Total 6.
2. Garden, carpet, tank. Analytic rubric: *Fencing* (0-2: 40 m); *Carpet* (0-2: 20 square metres); *Tank* (0-2:  $3.14 \times 1 \times 1 \times 2 = 6.28$  cubic metres = 6,280 litres); *Check and clarity* (0-2, **Analyze:** one estimation check, for example the tank as a  $2 \times 2 \times 2$  box = 8,000 litres, and steps laid out for re-checking). Total 8.
3. Book corner. Analytic rubric: *Carpet* (0-2: 6 square metres); *Skirting* (0-2: 5 m); *Shelves* (0-2:  $250 \times 2.5\text{ cm} = 6.25\text{ m}$ ;  $6.25 / 0.8 = 7.8$ , so 8 shelves); *Under-planning judged* (0-2, **Analyze:** a single 4-shelf bookcase leaves 4 shelves short, shown by the computation). Total 8.
4. Courtyard with store. Analytic rubric: *Skirting* (0-2:  $2 \times (30 + 20) = 100\text{ m}$ ; accepting that the store's notch adds or subtracts a few metres only if argued); *Tiles* (0-2:  $600 - 20 = 580$  square metres); *Re-measure analysis* (0-3, **Analyze:** the tiles depend on the open area, which changes with the store's position, so the tile quantity needs the re-measure - the boundary is unaffected by where the store sits inside it); *Units* (0-1). Total 8.
5. Practicum measurement plan. Analytic rubric: *Plan presented* (0-2: all four moves with units); *Weak move identified* (0-2, **Analyze:** the weakest move named with evidence); *Teacher's verdict reported* (0-2: what the cooperating teacher actually said); *Communication improvement* (0-2: a

concrete change that makes the plan independently re-checkable); *Clarity* (0-2). Total 10.

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